

Evidence-Gap Assessment: Architecture Definition and Management

Base Practice	Status	Rationale
Prepare for architecture definition	EvidenceStatus.NO_EVIDENCE_IDENTIFIED	There are no claims or evidence suggesting a systematic approach to preparing for architecture definition in terms of a roadmap, approach, or evaluation criteria. There is an absence of evidence regarding stakeholder analysis, enabling systems, or plans specific to architecture definition.
Develop architecture viewpoints	EvidenceStatus.PARTIAL_EVIDENCE	While development viewpoints and frameworks are inferred, evidenced by the use of Apache Arrow for its technical capabilities, there is no explicit rationale documented for selecting frameworks. Claims about decisions being influenced by primary investigators suggest a lack of formal viewpoint selection or modeling techniques.
Develop models and views of candidate architectures	EvidenceStatus.PARTIAL_EVIDENCE	Evidence suggests some level of modeling with emphasis on integrating clients with solver backends. However, there is limited detail on how models were composed or assessed in relation to stakeholder concerns or software requirements, especially with no requirements traceability mentioned.

Relate the architecture to design	EvidenceStatus.NO_EVIDENCE_IDENTIFIED	There is no evidence indicating a structured mapping and integration of architecture decisions into design elements. Without mention of interfaces or alignment between architecture and system elements, this remains unassessed.
Assess architecture candidates	EvidenceStatus.NO_EVIDENCE_IDENTIFIED	There is limited information on how architecture candidates were evaluated against stakeholder requirements or constraints, beyond informal processes led by principal investigators. No architecture assessment process or criteria is described.
Manage the selected architecture	EvidenceStatus.NO_EVIDENCE_IDENTIFIED	No evidence suggests an established governance approach or formal management of the architecture post-selection, with no acceptance from stakeholders documented. Claims imply a research-driven rather than governance-driven progression, lacking formal traceability or management artifacts.